Where the yield actually peaks

A field trial varies nitrogen and irrigation across plots. Yield saturates in both. Where is the optimum, and when water is rationed, how should the remaining input budget be split?

Abstract

A field trial varies nitrogen and irrigation across plots. Yield saturates in both. Where is the optimum, and when water is rationed, how should the remaining input budget be split? State the box the doses can live in. 7 steps are recorded in full below. The design was twelve plots and it recovered a two-treatment saturating surface well enough that the true curve sits inside the 90% band at every nitrogen dose. That is the case for spending plots on curvature rather than spreading them evenly: a 5x5 grid would have cost twice as much and learned the shape less well. No assumptions were recorded for this run, which is not the same as none being required.

1. Introduction

This report addresses one question: A field trial varies nitrogen and irrigation across plots. Yield saturates in both. Where is the optimum, and when water is rationed, how should the remaining input budget be split?

2. Methods

Question: A field trial varies nitrogen and irrigation across plots. Yield saturates in both. Where is the optimum, and when water is rationed, how should the remaining input budget be split?

State the box the doses can live in

Bounds are not a plotting range. They are a claim about the world — no plot on this farm will receive 400 kg/ha of nitrogen, so no optimizer should ever propose it and no fitted curve should be trusted out there. Saying it once, in an object, means the design, the allocation and the frontier all obey the same box instead of each carrying its own copy of the farm's limits.

Considered instead: Fitting on whatever doses happened to occur and reading the optimum off the curve. That silently extrapolates: a Hill curve is still increasing at every dose, so the unconstrained answer to 'how much nitrogen' is always 'more'.

```
treatments : ('nitrogen', 'irrigation')
nitrogen : 0 to 200 kg/ha
irrigation : 0 to 120 mm
```

Spend the plots where the curvature is

Curvature is what this whole analysis turns on, and curvature is estimated from runs placed to see it bending. A rotatable central composite design is a factorial cube, plus axial points pushed out along each treatment, plus replicated centre points. The axial points are what make the quadratic terms estimable; the replicated centre is what separates pure noise from lack of fit. 'Rotatable' means the prediction variance depends only on distance from the centre, so no direction in the dose box is privileged by accident.

Considered instead: A 5x5 grid over the same box. It costs 25 plots instead of twelve, spends most of them re-measuring the flat regions, and still gives no replicated centre — so it cannot tell a bad model from a noisy one. One factor at a time is worse again: it cannot see an interaction at all.

```
design : central_composite, 12 plots
detail : {'alpha': 1.4142135623730951, 'n_cube': 4.0, 'n_axial': 4.0,
'n_center': 4.0, 'fraction': 0.0}
```

nitrogen	irrigation
29.3	17.6
170.7	17.6
29.3	102.4
170.7	102.4
0	60
200	60
100	0
100	120
100	60
100	60

100	60
100	60

Table 1. Every plot the trial will plant, in the units an agronomist works in

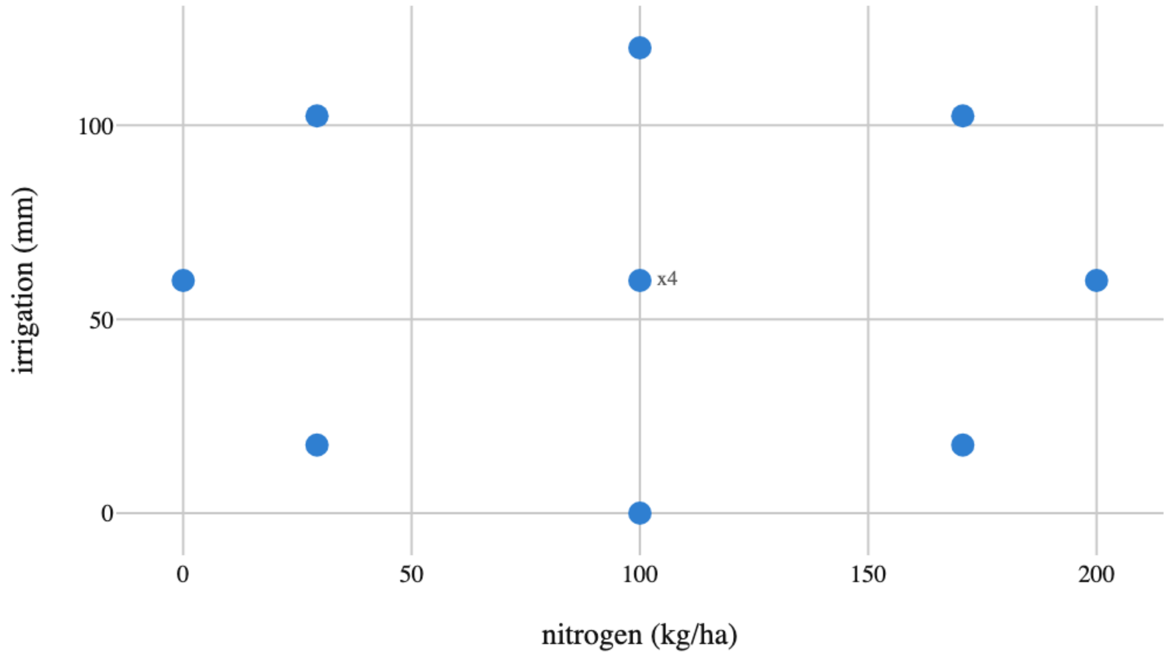

Figure 1. The twelve plots, in dose space. Four corners, four axial points pushed out along each treatment, and a centre planted four times over — that replication is the only thing in the design that can separate noise from a wrong model

Grow a field whose answer is already known

Every number below is checkable only because the truth was written down first. A recovery test is the difference between a method that works and a method that has never been graded: simulate from a known set of parameters, fit, and ask whether the intervals contain what generated the data.

Considered instead: A real dataset. It would be more convincing about agronomy and completely uninformative about whether the estimator is right — with real data there is nothing to compare the answer to.

```
plots : 12
true peak : nitrogen k = 90, irrigation k = 45
noise sd : 0.35 t/ha, against a yield range of about 3 to 9
```

Fit once, and check the truth is inside the interval

The likelihood calls exactly the same forward() the allocation and the frontier will call later. That is a design rule rather than an efficiency note: the moment the optimizer gets its own copy of the transform chain, the two drift, and the drift shows up as a recommendation the model cannot actually justify.

Considered instead: Reporting the point estimates and moving on. The comparison worth making is not 'is the mean close' but 'does the interval contain the truth' — an estimator can be nearly right and still badly overconfident, and only the second question catches that.

```
converged : True
parameters: alpha, beta_irrigation, beta_nitrogen, k_irrigation,
k_nitrogen, s_irrigation, s_nitrogen, sigma
```

parameter	truth	posterior mean	90% HDI	contains truth
beta_nitrogen	4.2	5.1	[3.9, 6.3]	yes
k_nitrogen	90.0	108.2	[75.6, 137.4]	yes
beta_irrigation	2.6	2.9	[1.6, 4.1]	yes
k_irrigation	45.0	76.4	[43.6, 106.2]	yes

Table 2

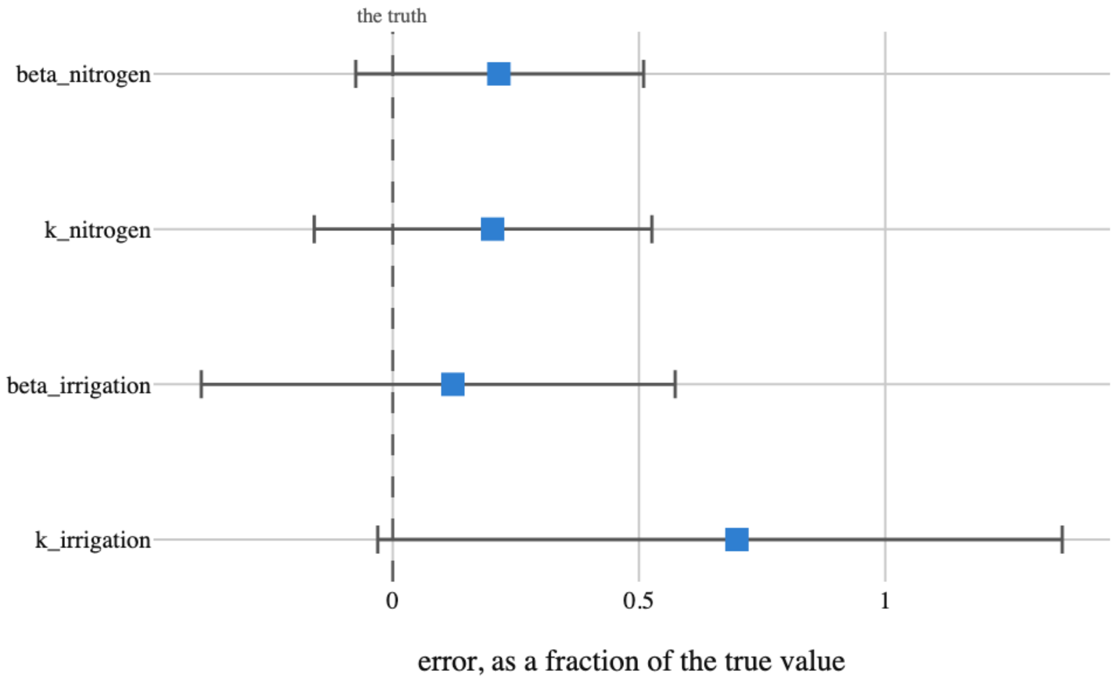

Figure 2. Parameter recovery, every parameter on one axis. Each interval is scaled by its own true value, which is the only way to put a half-saturation dose in kg/ha and an amplitude in t/ha on the same picture. An interval that misses the dashed line is a recovery failure and would be drawn in red

Ask where the response flattens out

The band is not a fitted line with error bars bolted on. It is every posterior draw pushed through forward() and summarised at each dose, so the width at a given dose is what the model genuinely does not know there — widest where the design placed fewest plots, which is exactly where a recommendation should be least confident.

Considered instead: Reading the optimum off the posterior mean curve alone. The mean crosses its own plateau somewhere; the band shows a whole region of doses the data cannot tell apart, and that region, not the point, is the honest answer to 'how much nitrogen'.

nitrogen (kg/ha)	posterior mean (t/ha)	90% interval	truth
0	4.33	[4.08, 4.58]	4.40
28	4.88	[4.71, 5.09]	4.76
55	5.67	[5.50, 5.88]	5.54
83	6.38	[6.24, 6.57]	6.32

110	6.95	[6.82, 7.12]	6.92
138	7.38	[7.26, 7.54]	7.34
166	7.70	[7.56, 7.87]	7.64
193	7.96	[7.76, 8.15]	7.85

Table 3. Every fourth grid point; the chart below has all thirty

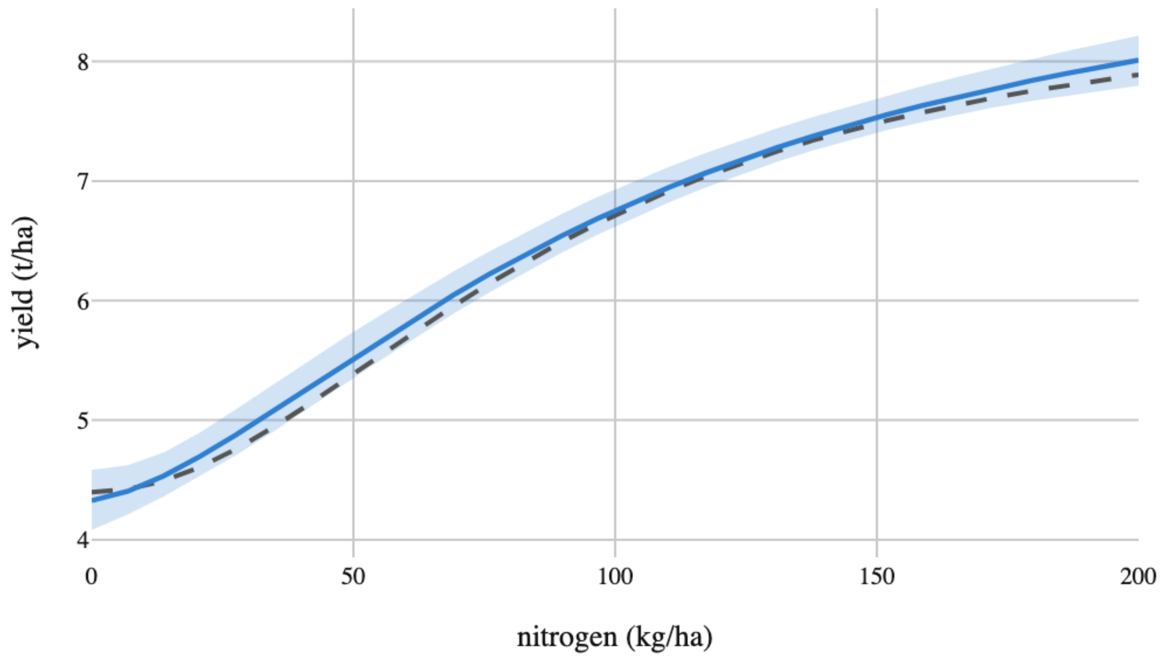

Figure 3. Yield against nitrogen, with everything the model does not know. The dashed line is the curve the simulation actually used. It stays inside the band across the whole dose range, which is what a passed recovery test looks like when you draw it instead of tabulating it

Now ration the water and split the budget

The decision is not 'where is the peak' — it is 'given a fixed total of input and a water cap, what mix'. allocate() searches the fitted surface under both constraints, using the posterior rather than a single fitted curve, so the recommendation reflects the uncertainty in the shape and not just its mean.

Considered instead: Splitting the budget in proportion to each input's estimated amplitude. That is the right answer only for a linear response; with saturation it over-buys whichever input is already past its bend, which is precisely the mistake the surface layer exists to prevent.

```
budget 180 units of input, water capped at 60 mm:
nitrogen 120.0, irrigation 60.0
expected yield 7.184 t/ha (converged)
the same budget with no water cap:
nitrogen 106.3, irrigation 73.7
expected yield 7.229 t/ha
the cap costs 0.045 t/ha
```

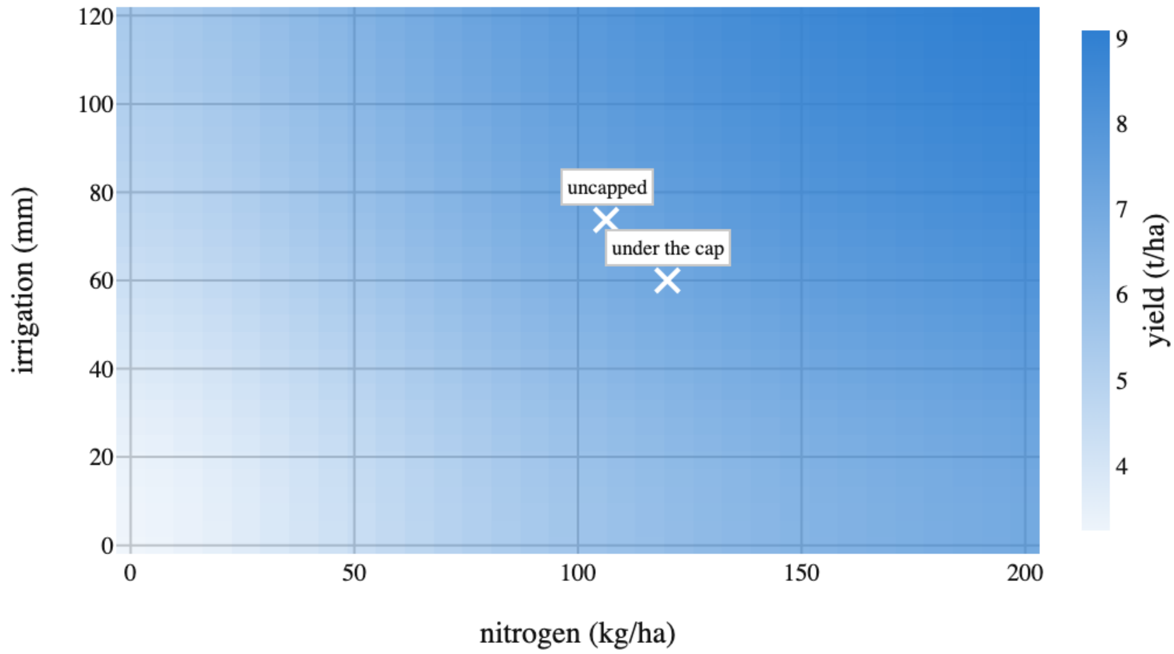

Figure 4. The whole fitted surface, and the two allocations. Both marks sit on the same budget line — 180 units of input — and differ only in whether irrigation may exceed 60 mm. The surface is flat enough along its ridge that the cap costs surprisingly little

Price the next unit of input

A planner cannot act on 'the optimum'. They can act on 'the next hundred units of input are worth this much yield, the hundred after that are worth this much less'. The frontier re-solves the allocation at each budget, and the shadow price is the slope between neighbouring solutions — the number that says when to stop buying.

Considered instead: Quoting one average return per unit of input. On a saturating surface the average is a blend of a steep early region and a flat late one, so it overstates the value of the next unit whenever you are already past the bend — and understates it when you are below.

budget	expected_outcome	shadow_price	nitrogen	irrigation
60	4.654	0.022	60	0
120	6.002	0.02	120	0
180	7.079	0.016	120	60
240	7.862	0.008	180	60
300	8.041	0.003	200	60

Table 4. The best achievable yield at each budget, and how it is spent

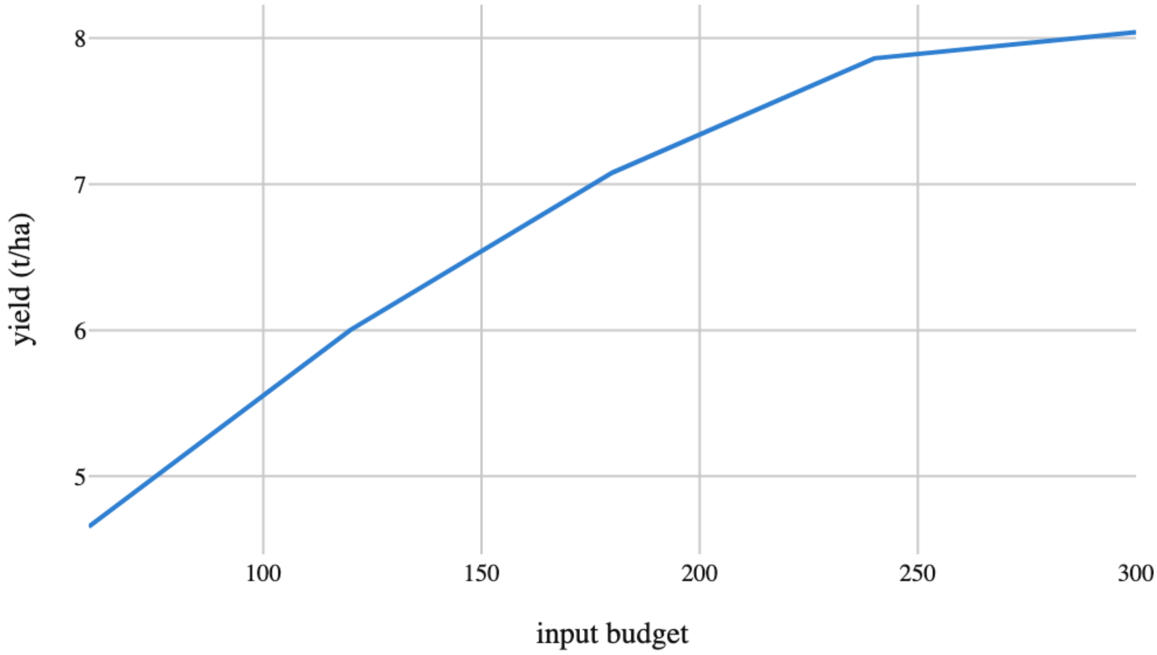

Figure 5. What each budget buys. The curve bends: equal increments of budget stop buying equal increments of yield

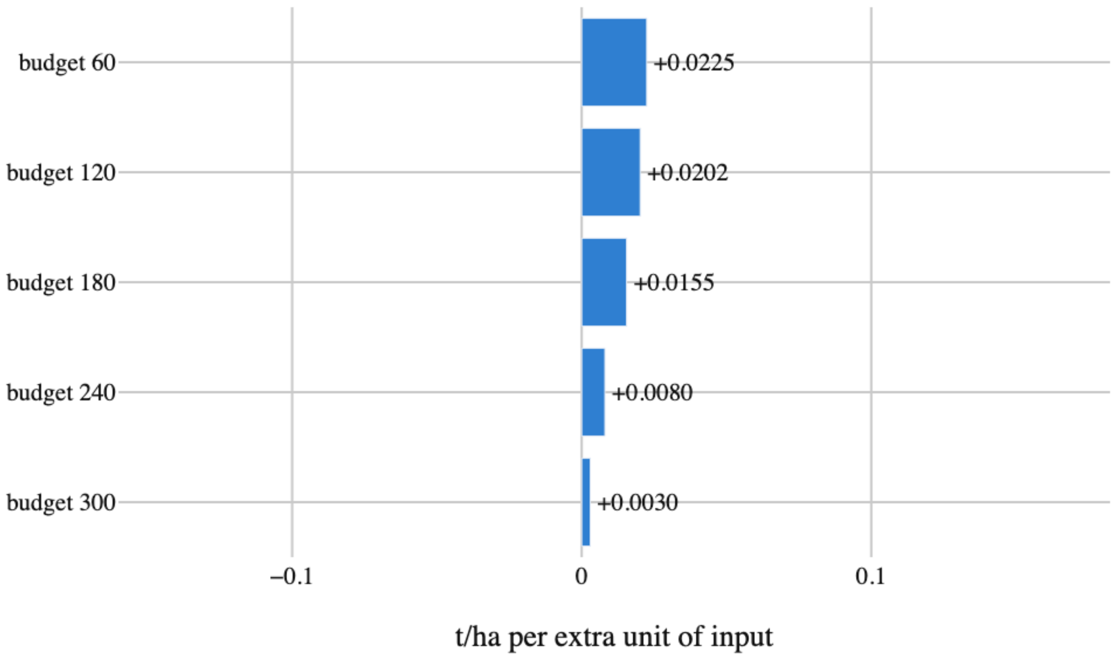

Figure 6. The shadow price of the next unit, by budget. This is the frontier's slope, and it is the number a budget conversation is actually about. It falls monotonically, and the budget worth funding is the largest one whose shadow price still beats the price of the input

3. Results

No findings were recorded.

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

The design was twelve plots and it recovered a two-treatment saturating surface well enough that the true curve sits inside the 90% band at every nitrogen dose. That is the case for spending plots on curvature rather than spreading them evenly: a 5x5 grid would have cost twice as much and learned the shape less well.

The water cap turned out to be cheap. Capping irrigation at 60 mm costs 0.045 t/ha at a budget of 180 — the surface has a broad ridge, so the constraint slides the answer along it rather than down it. Had the ridge run the other way the same cap would have been expensive, and nothing but the fitted shape tells you which case you are in.

The shadow price falls from 0.0225 to 0.0030 t/ha per unit across the budgets considered — a factor of 7.6. A single average return per unit of input would have hidden that entirely, and it is the only number in this whole run that answers the question a planner actually asked: should we buy more?

5. Model checking

No diagnostics were recorded. An analysis with no model checking is not therefore sound; it is unexamined.

6. Discussion

There are no findings to interpret.

7. Conclusions

A field trial varies nitrogen and irrigation across plots. Yield saturates in both. Where is the optimum, and when water is rationed, how should the remaining input budget be split?

No finding was recorded, so no conclusion follows.

8. Limitations

No assumption in the record is left unresolved. That is a statement about the record, not a guarantee about the world.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Field	Value
field	Agronomy
source	examples/ walkthrough record
steps	7
evidence hash	e353f63afc4b871d31dc1eed8c74cab1ff23ec418dd612364500a101e923da66

Table 5. Run